MACRO RESEARCH MEMORANDUM

What the Yield Curve Is Currently Signaling: A Framework-Based Assessment

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Model basis: Nelson-Siegel yield curve decomposition + logistic regression recession model, 1990–present daily US Treasury data. Out-of-sample ROC-AUC: 0.756.

The conventional view is that the yield curve inversion of 2022–2024 was a false signal — the soft landing succeeded, the Fed navigated perfectly, and the danger has passed. I think this is wrong. Not confidently wrong, but probably wrong in a way that matters for anyone allocating across rates, credit, and equities right now.

I. THE MECHANISM: WHY INVERSIONS PRECEDE RECESSIONS

Yield curve inversions do not cause recessions. They are a symptom of a specific set of conditions that reliably produce them. The causal chain runs as follows.

When the Federal Reserve tightens monetary policy aggressively, it raises the federal funds rate — a short-term rate. The 3-month Treasury yield tracks this directly. Long-term yields (10-year, 30-year) are set by the bond market and reflect what investors expect for long-run nominal growth and inflation. When the Fed overtightens — raising short rates above what long-run growth can support — the short end of the curve rises above the long end. The curve inverts.

Banks make money by borrowing short and lending long. Their net interest margin is the spread between what they pay depositors (short-term rates) and what they earn on loans (long-term rates). When the curve inverts, that margin collapses or turns negative. Banks stop lending. Businesses cannot fund expansion. Consumers cannot access credit for housing and durables. The credit contraction propagates through the real economy with a lag of 12 to 18 months — which is why the yield curve is a leading indicator, not a coincident one.

This mechanism is not a statistical curiosity. It is a cause-effect chain embedded in the structure of fractional reserve banking. The Nelson-Siegel slope factor (β_1) from my model quantifies this: in expansion periods, β_1 averages +2.8%; in recession periods it averages +0.4% — a full 240 basis point compression that precedes the NBER declaration, not follows it.

II. WHAT THE DATA SHOWS

I built a logistic regression recession prediction model using 34 years of daily US Treasury data — five maturities from 3-month to 30-year — decomposed via Nelson-Siegel factor modeling. Three findings define the framework.

Finding 1: The model works. Using only yield curve features — the 2s10s and 3m10s spreads, the Nelson-Siegel slope and level factors, and the 3-month momentum of curve flattening — the logistic model achieved an out-of-sample ROC-AUC of 0.756 on post-2010 data it had never seen during training. For context, a random model scores 0.50 and the target threshold was 0.70. The yield curve contains genuine forward-looking information about recession risk.

Finding 2: The most dangerous moment is the re-steepening, not the inversion. In four of the five meaningful inversions since 1990, the curve had already begun re-steepening — often for three to six months — by the time the recession was officially underway. The 2s10s spread went from -0.52% at its most inverted in 2006 to slightly positive by early 2008, right as the financial crisis was materializing. The re-steepening reflects the Fed cutting in response to emerging weakness, not the absence of weakness. It is stress arriving, not stress departing.

Finding 3: Asset class behavior validates the mechanism. Across 2003–present, backtesting shows that long-duration Treasuries (TLT) returned -2.1% annualized in inverted curve regimes — because the inversion is caused by the rate hikes that destroy bond prices — but historically recover sharply when the Fed pivots. High yield credit (HYG) returned 4.0% in inverted regimes versus 5.5% in steep regimes, with meaningfully wider drawdowns, as credit spreads anticipate default risk. Equities (SPY) showed a Sharpe ratio of 0.55 in steep regimes versus 0.93 in inverted regimes — elevated short-term returns as stocks initially shrug off the inversion signal, followed by the worst drawdowns in the dataset.

III. CURRENT MARKET ASSESSMENT

As of April 2026, the 2s10s spread sits at approximately +0.55%, having re-steepened from a trough of -1.07% in mid-2023 — the deepest inversion in the dataset's 34-year history. My model currently places 12-month recession probability at approximately 50–60%, above the 43% decision threshold, despite the curve having nominally normalized.

The consensus reads the re-steepening as resolution. I read it as the most historically dangerous phase of the cycle. The 2022–2024 inversion was deeper and longer than any prior episode in my dataset. The credit contraction it initiated — commercial real estate, regional bank stress, tightening consumer credit standards — has not fully worked through the real economy. The Fed has begun cutting, which is historically what happens when weakness is already visible in data that lags the underlying conditions by 3 to 6 months.

The implication for positioning: long duration fixed income is not the trade during inversion — it is the trade after the pivot is confirmed and the curve steepens aggressively as cuts accelerate. Credit spreads at current levels price a soft landing scenario I assign roughly 35% probability. Equity valuations embed earnings growth assumptions that require either a productivity acceleration I cannot yet see in the data or a Fed that cuts faster than current pricing implies. The asymmetry favors being earlier on duration and more cautious on credit spread risk than consensus positioning

currently reflects.

IV. WHAT WOULD MAKE THIS VIEW WRONG

I hold three falsifying conditions seriously, and I want to be precise about what each would require to actually overturn the thesis.

First, a structurally higher neutral rate. If r* has genuinely shifted upward post-pandemic — due to fiscal deficits, deglobalization, or defense spending — then current Fed policy may be less restrictive than the historical comparison suggests. An inverted curve at a 4% neutral rate is a different signal than an inverted curve at a 2% neutral rate. I find this plausible and assign it meaningful weight, but note that bank lending standards, commercial real estate delinquencies, and credit card stress data do not currently suggest conditions consistent with a non-restrictive policy environment.

Second, fiscal offset at unprecedented scale. Post-pandemic fiscal transfers and the Inflation Reduction Act represent direct demand injection that can bypass the credit transmission mechanism entirely. If the government is spending enough to offset private credit contraction, the inversion's historical predictive power is diminished. This is the most empirically credible challenge to my view. I watch fiscal impulse data carefully as the primary disconfirming signal.

Third, AI-driven productivity acceleration. Genuine supply-side growth could produce real output expansion that makes the demand weakness from tight credit irrelevant. I find this the least convincing falsifier in the near term: productivity gains diffuse across the economy over years, while credit conditions tighten within quarters. The timeline mismatch means AI would need to be generating measurable economy-wide productivity gains now — not in 2027 — to offset the current cycle's dynamics.

My honest probability distribution: soft landing 35%, delayed recession 45%, genuinely novel outcome the historical data cannot speak to 20%. The 80% soft landing confidence embedded in current market pricing is, in my view, overconfident by a factor that should matter to anyone managing duration, credit, or equity volatility exposure across the next 12 months. The question is genuinely open. The mechanism is underappreciated. The complacency itself is the risk.

Model: Nelson-Siegel yield curve decomposition (1990–present) + logistic regression recession predictor. Data: FRED (DGS3MO, DGS2, DGS5, DGS10, DGS30, USREC) + yfinance (SPY, TLT, HYG, LQD). Out-of-sample evaluation period: 2010–present. ROC-AUC: 0.756. Full methodology and code: github.com/kanavseth/yield-curve-analysis